

HiveForce Labs

THREAT ADVISORY

 **ATTACK REPORT**

ValleyRAT's Stealthy Job-Lure Campaign

Date of Publication

December 5, 2025

Admiralty Code

A1

TA Number

TA2025370

Summary

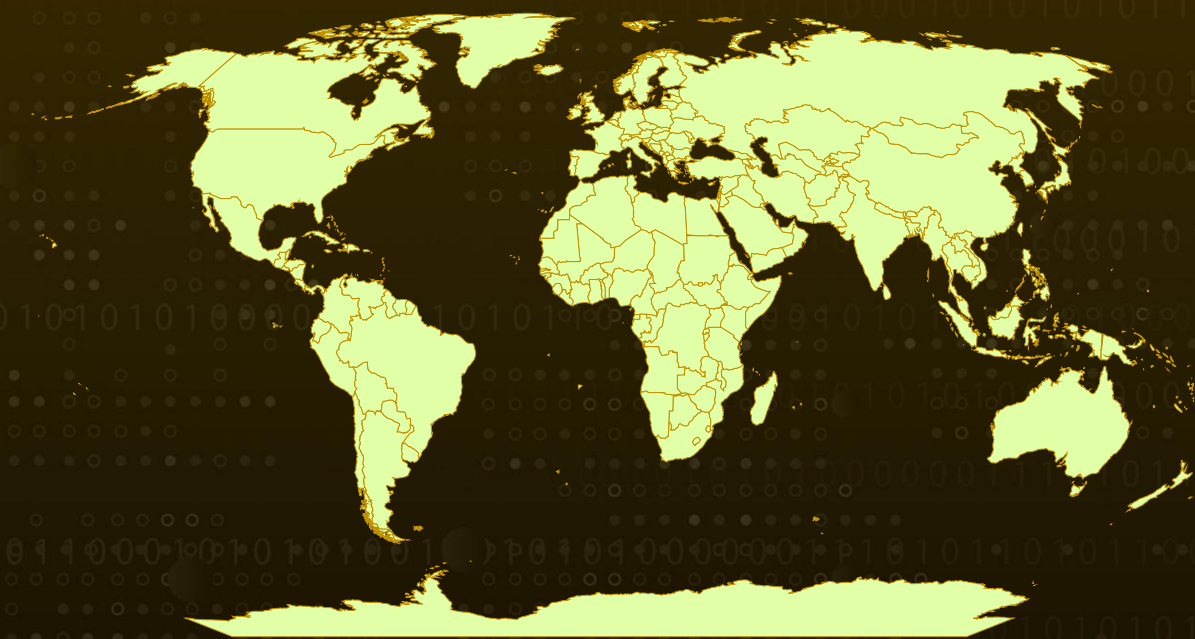
Attack Discovered: End of October 2025

Targeted Region: Worldwide

Malware: ValleyRAT

Attack: ValleyRAT's latest campaign cleverly preys on job seekers by disguising malware as convincing recruitment documents, using familiar names, trusted logos, and layered technical tricks to slip past defenses. By abusing Foxit software, hiding malicious DLLs, and bundling a portable Python environment, the attackers create an infection chain that runs quietly in the background while victims believe they're opening harmless job files. The operation ultimately steals browser data and maintains persistence through stealthy registry modifications, all while masking its activity. This campaign highlights how emotional vulnerability, like the urgency of finding a job, can be just as powerful a tool for attackers as the malware they deploy.

Attack Regions



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Attack Details

#1

Cybercriminal activity continues to grow more aggressive and refined, with attackers blending social engineering, heavy obfuscation, and DLL sideloading to improve their chances of success. A recent campaign illustrates just how effective this layered approach can be, marked by a notable spike in [ValleyRAT](#) detections. What began as an activity aimed primarily at Chinese-speaking users has now broadened to include English-speaking job seekers, an audience more likely to trust documents disguised as hiring materials.

#2

Job seekers, often optimistic and anxious for new opportunities, can easily drop their guard when presented with convincing recruitment files. Threat actors capitalize on this by using filenames such as `Overview_of_Work_Expectations.zip` and `Candidate_Skills_Assessment_Test.rar`, which appear routine in the hiring process. These files conceal malicious components behind the façade of HR documentation, making unsuspecting users far more likely to open them.

#3

A central element of the campaign is the misuse of Foxit software. Attackers embed a renamed `FoxitPDFReader.exe` within their malicious archives, presenting it as a legitimate hiring document, sometimes under names like `Compensation_Benefits_Commission.exe`. By replicating the Foxit logo, they lead victims to believe they are opening a simple PDF. The executable silently loads malicious code in the background, enabling ValleyRAT to run while the user focuses on the decoy content.

#4

The infection chain itself is intricate. The disguised `FoxitPDFReader.exe` loads a malicious `msimg32.dll`, which is crucial for triggering the ValleyRAT payload. The archive also contains hidden directories and files intended to enhance the illusion of legitimacy, including a “Document” folder filled with innocuous-looking subfolders and disguised LNK shortcuts. A batch script, `document.bat`, uses a doctored `document.docx` file that also serves as a 7-Zip executable to extract `document.pdf`, which contains an embedded Python environment. This ensures the malicious script executes even if Python is not already installed on the system.

#5

From there, the attackers run an encoded Base64 Python script that serves as a shellcode loader, contacting an external IP to retrieve the final payload. The Python interpreter itself is renamed to avoid suspicion, and persistence is established via an autorun registry entry. The malware ultimately focuses on harvesting personal data from the victim’s web browsers. Network analysis of the captured certificates reveals common hallmarks of AsyncRAT, like infrastructure, self-signed certificates, random common names, deprecated TLS protocols, and unusually long validity periods, pointing to the automated tooling often used in commodity RAT operations. A campaign like this serves as a reminder that in today’s threat landscape, staying vigilant and questioning even the most convincing files is essential to keeping both personal data and systems secure.

Recommendations



Be Cautious with Unexpected Job-related Files: If you receive a ZIP, RAR, or EXE file claiming to be a job description or assessment, double-check the sender. Legitimate recruiters rarely send executable files. When in doubt, ask them directly.



Look Closely at File Extensions and Icons: A file may look like a PDF, but it could be an EXE hiding behind a familiar logo. Always hover over the file or check its properties before opening anything unfamiliar.



Use Multi-factor Authentication (MFA): If attackers manage to steal your browser data or credentials, MFA can prevent them from accessing your accounts.



Avoid Running Files That Require You to “Enable Content” or Install Additional Components: Documents that ask for permissions, scripts, or extra software are major red flags and should be treated with skepticism.



Enhance Endpoint Protection: Deploy next-generation antivirus (NGAV) and endpoint detection & response (EDR) solutions to identify and block malware. Leverage behavioral analysis and machine learning-based detection to spot suspicious activity.



Potential MITRE ATT&CK TTPs

<u>TA0001</u> Initial Access	<u>TA0002</u> Execution	<u>TA0003</u> Persistence	<u>TA0004</u> Privilege Escalation
<u>TA0005</u> Defense Evasion	<u>TA0006</u> Credential Access	<u>TA0007</u> Discovery	<u>TA0010</u> Exfiltration
<u>TA0011</u> Command and Control	<u>T1566</u> Phishing	<u>T1566.001</u> Spearphishing Attachment	<u>T1059</u> Command and Scripting Interpreter
<u>T1059.003</u> Windows Command Shell	<u>T1059.006</u> Python	<u>T1204</u> User Execution	<u>T1204.002</u> Malicious File

<u>T1547</u> Boot or Logon Autostart Execution	<u>T1547.001</u> Registry Run Keys / Startup Folder	<u>T1574</u> Hijack Execution Flow	<u>T1574.001</u> DLL
<u>T1027</u> Obfuscated Files or Information	<u>T1036</u> Masquerading	<u>T1036.005</u> Match Legitimate Resource Name or Location	<u>T1140</u> Deobfuscate/Decode Files or Information
<u>T1083</u> File and Directory Discovery	<u>T1555</u> Credentials from Password Stores	<u>T1555.003</u> Credentials from Web Browsers	<u>T1071</u> Application Layer Protocol
<u>T1071.001</u> Web Protocols	<u>T1041</u> Exfiltration Over C2 Channel		

⌘ Indicators of Compromise (IOCs)

TYPE	VALUE
SHA1	ebcfc4f6c6e63b75dc407f5e76c9d96c69c3c1b6, 5cb888c87b15ec998c638892ad382dc68efb7f94, 65fec70eaca638cbd10a6774e4e67f2d55f63959, 9eb12480a9e3be552c88960d45beeacfb3b2444b, 0227738e5a98622ea88a2f09527618a6fc4b9be9, 37a9972e667dc479bbbd586803d809b56b00ffcf, a2f95e24cb46a0a1edba7d1ff4f7fa2789ee2325, c52473f4e858e91dfd769a1774e5821750b8c8e2, 9857742e8f10595111be8cf813f4b83c6e8d7bc5, da24feaa3f0fc9a326905afab295324944fb0f03, 678045a8b05e4bf36c7bb0338612bcbfc2376dd8, 3776c0df00a24f9152b6cb44c0fa05ef410f747b, 191a7e68ca0d15af4283844aa4916f8f73fe6be2, 2ed9043c9eaf0cccb96711ca05a07add8a10ec88, e43daace2965439f2b1574b903d08af427f9307d, 21f6eb9e582043af0be06dbdb2819e9642ca991a, 1c796a576e1f30ed0a672b30b0d2644aeb7f739, 7dab6af4c90599ddba656bec8e258d08e9e8ffe6, df86e5d1a2d614c762704dc8842874fad27de978, 2afde2c2a7773f68de691fe4f4e835fe530a3471, 784468f153fd9b1676504f51bce1bbd9237373f2, 2336179beb8a8d4c9a23771cc0ae31f2a146fbbd, 763b0273c03053405806722df9fe3c9e270c8d55, de263d9374d90fb7f51b19a036cb75a3d2072342

TYPE	VALUE
SHA256	a32fa6ba08db96ebd611f6ee06da44b419d569a6bac43ed00c68d6ca674004c3, 7e8415e2744be160b7d7c600a401de41554c1357c2d2d35c85f8be8068cbc649, 9b0afe79696ccb263b8a00c75c021d115f152283714c0e4c5075aad4e52b94f9, 074400e2f09312081c29e905a9d24f70cfc5f535cd1dcaaca31e33586c7b01bc, cabd71a7a4df7fa6b5ffe0f22354953b5d278c5b2626af8bfba0ba726acc59ef, 3b2d397f308e00f6ef5ae4a368b6ec9a1b5791b883583d104949d80711f5789f, c3f09771a248daede16382ec9484c6a626e2f289c095164eea97170f3e4a6769, 8a18b8826daafc4a84d49299013c5cabe95dd9159ea5d5f1fb5872a6d70666e6, 28fb1c360663b13a4f918b76a12bdc6f7532896eabf8200bddb63319c92bad26, ff84bb121533144fc87c314a9d50c16dd15bb7d8f036c777e0a6c1dc7395e000, 99af1fe7e00d4d82bc6ae4440ad3528202a8a6234038fcde15e78dfea79dac2a, 6266d87b93e8129b0b606971f0c9e00214abadbc758769bd9cf456c6e0ad8b6e, e65b359519b912139ae7ab3ac77c667c5411d2264a1d75166ae2dffffe2efaa, 0010c5caa8311201cd3b0e335a3936e7d1143362d98f4b5a57ef780dcdc1ca5a, 8046fe163a0ab581df7ea7c86788d7dca42f70fac95023dfb36d9281ad3463d9, 2261efb7516dd49edd3bface0c769a531c37ce0ca6832871768f622abb0f1f71, cb30d5b932a461601deaf2ef76476e216c7d2a99ea7c280cadbe6510b2997080, 6ec18bf62078bb2661b2d0cadf0314ea44fc67da786c28456869b0102eea235c, 6351649b644eafe5c27317251bd466a8eab265defb58ab34c20feec60304a861, f75c4d6c989c03c339ead7708227f22d30b9a5f4f433bbe29391a1003aa85d85, bb21ec0bb2b94c5471ed7c768cc999808a42e985955384f6af360caa0c640d6c,

TYPE	VALUE
SHA256	216cd0ccc129a612082bddf2502a58aef6b1c22ea07a18a58e4e8315d6e a3fbb, f49ba5d85a7be63599346097278c1af49ab6c1bea82e422462057c78641 d54d4, ce1be9e4b2fd0f3958720f9bc2ae9d545bc0e27dcf1042b64a70f1fc62884 610
URLs	hxxp[:]//196[.]251[.]86[.]145/huna, hxxp[:]//51[.]79[.]214[.]125/huna
IPv4:Port	154[.]90[.]58[.]164[:]56001
Filenames	Overview_of_Work_Expectations.zip, Candidate_Skills_Assessment_Test.rar, Authentic_Job_Application_Form.zip

References

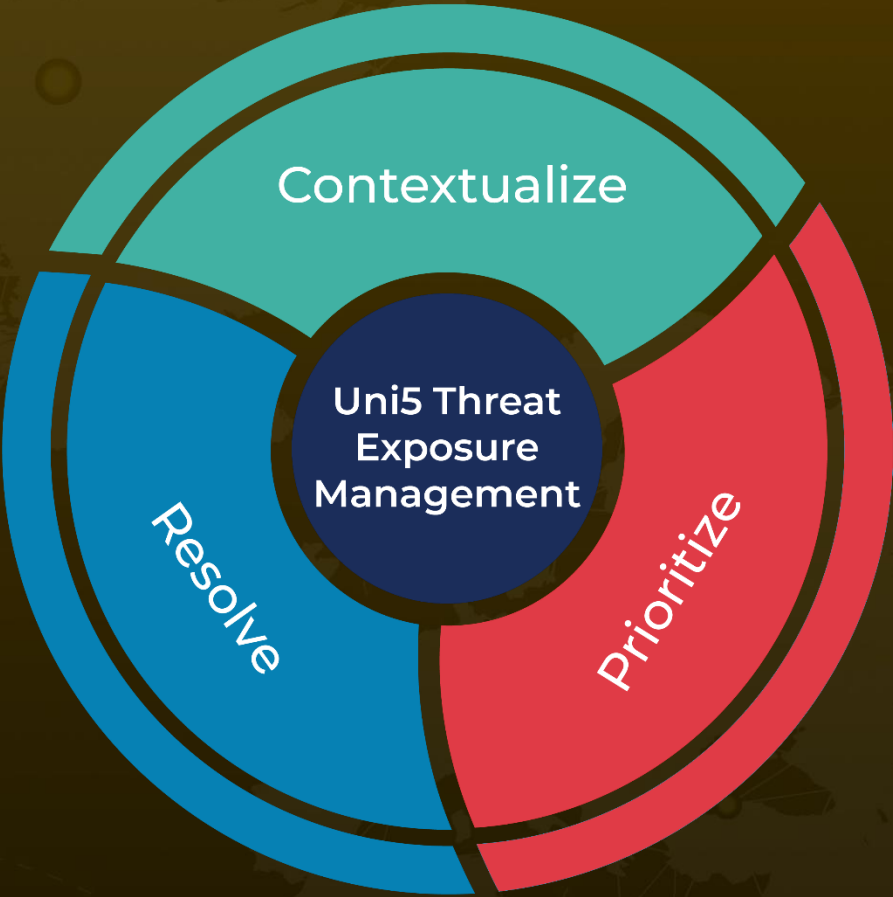
https://www.trendmicro.com/en_us/research/25/l/valleyrat-campaign.html

<https://hivepro.com/threat-advisory/operation-silk-lure-scam-when-job-hunts-leads-to-malware/>

What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

Book a free demo with HivePro Uni5: Threat Exposure Management Platform.



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